Original Article

Biofpr Biofuels Bioproducts & Biorefining

Optimization of glycerol consumption in wild-type *Escherichia coli* using central carbon modeling as an alternative approach

Albert Enrique Tafur Rangel, Grupo de Diseño de Productos y Procesos, Department of Chemical and Food Engineering, Universidad de los Andes, Bogotá, Colombia; Grupo de Investigación CINBIOS, Department of Microbiology, Universidad Popular del Cesar, Valledupar-Cesar, Colombia
Luis H Reyes, **Jorge Mario Gómez Ramírez**, **Andrés Fernando Gónzalez Barrios PhD Associate Professor**, Grupo de Diseño de Productos y Procesos, Department of Chemical and Food Engineering, Universidad de los Andes, Bogotá, Colombia

Received December 3 2020; Revised January 12 2021; Accepted February 1 2021;
View online at Wiley Online Library (wileyonlinelibrary.com);
DOI: 10.1002/bbb.2205; *Biofuels. Bioprod. Bioref.* (2021)

Abstract: The development of a culture medium is an essential step in any bioprocess involving microorganisms for the bioconversion of by-products to valuable chemicals, making industries like the biofuel industry more competitive. Optimization of the bioconversion process to minimize cost while maximizing yield underscores the importance of using computational methods to identify cellular requirements under specific growth conditions. In this study, a computational approach was proposed as an alternative to optimizing glycerol consumption in one of the most common production chassis, *Escherichia coli*, specifically strain ATCC 8739. Nineteen compounds were identified as essential for *E. coli* growth in glycerol. Of these, three reactions associated with nitrogen, phosphorous, and oxygen availability were determined as crucial to reaching high growth and glycerol uptake rates. Based on computational results, a glycerol-based medium was supplemented with reported common chemical compounds that contain nitrogen or phosphorous (NH_4Cl, Na_2HPO_4, and K_2HPO_4) for further experimental validation. When comparing the supplemented culture medium experimentally with LB medium (Luria Bertani), a two-fold increment in the glycerol consumption was observed. Transcriptomic analysis of the most promising culture medium reveals that high glycerol utilization under aerobic conditions is dependent on phosphorus to avoid toxicity within the cell because of glycerol-3-phosphate generation. The result of this study is a resource to determine nutritional requirements that allow the improvement of the use of raw material for the production of compounds that are attractive to the bio-based industry. © 2021 Society of Chemical Industry and John Wiley & Sons, Ltd

Key words: optimization; culture medium; glycerol; *Escherichia coli*; genome-scale models

Correspondence to: Andrés Fernando Gónzalez Barrios, Grupo de Diseño de Productos y Procesos, Department of Chemical and Food Engineering, Universidad de los Andes, Bogotá, Colombia. E-mail: andgonza@uniandes.edu.co

Introduction

Petroleum is neither renewable nor ecofriendly but currently it is the most crucial energy source on the planet.[1] Eco-friendly and renewable alternatives such as solar, hydraulic, wind, and biomass are useful for meeting continuously growing global energy demands.[2] Biomass-based technologies have continuously grown and have resulted in the generation of biofuels. Nonetheless, during the bioconversion process of biomass to biofuels, numerous by-products are generated.[3,4] The generation of glycerol has become a burden, diminishing the industry's competitiveness.[5–8] Consequently, glycerol conversion into novel or valuable products such as polymers, surfactants, solvents, and chemical intermediates has garnered growing interest, and it is a promising alternative.[4,9]

Glycerol has become attractive for bio-based chemical production, mainly for two reasons. First, several microorganisms have been reported to have the ability to use glycerol for growth.[4,10] Second, glycerol is more highly reduced than sugars (e.g., glucose), making the production of reduced chemicals such as organic acids, ethanol, and hydrogen possible. The presence of the tricarboxylic acid cycle (TCA) and other microorganism pathways allows the production of organic acids from glycerol by using them as microbial cell factories, although naturally in low yields.[11–13] Organic acids, usually generated from petroleum, have been considered critical building-block chemicals because they have an attractive market and a wide range of applications.[11,13] The potential market for these acids may therefore be expanded if renewable raw sources are used, and the costs associated with glycerol's bioconversion are minimized at the same time as maximizing organic acid yields, making biodiesel production more economical and competitive.

For any microorganism to be used in biotechnological processes, the cells must be supplied with a biochemically appropriate extracellular environment. *Escherichia coli* has been used extensively for the industrial production of several bio-based organic acids because of its fast growth rate, known genetic background, and the many viable genetic tools developed to facilitate its study. However, whether this behavior can be used as an efficient microbial cell factory strongly depends on its growth environment. In *E. coli*, low carbon source consumption / uptake has been observed when simple or enrichment culture media are used in both wild-type and engineered strains, increasing the cost of fermentation.[4,14–17] Enhancing glycerol uptake via genetic manipulation or culture media optimization is a critical step in the process of biochemically converting glycerol into more useful chemical products at reduced costs.[18]

Three strategies can be considered to enhance the consumption and uptake rate of glycerol: (i) experimental-based culture design, (ii) adaptive laboratory evolution (ALE), and (iii) systems metabolic engineering. The first strategy involves using an experimental design to identify optimal concentrations and composition of specific compounds that maximize the desired primary or secondary metabolites' production. This technique has been used widely to produce target compounds by testing an extensive range of concentration compounds of culture media, culture media itself, and incorporating additives.[19–21] These types of experiments involve a cumbersome, labor-intensive, expensive, and tedious process.[20,21] Adaptive laboratory evolution is a powerful technology, which takes advantage of (i) the rapid and consistent growth, (ii) large population sizes, and (iii) capacity to be stored via freezing for subsequent examination for developing both compound-specific and platform-based bioprocessing strains.[16,22] During ALE, a microorganism is cultivated under clearly defined conditions for prolonged periods in the range of weeks to years, which allows for the selection of improved phenotypes. [22–25] The use of adaptation to facilitate growth or uptake rate improvements is especially relevant concerning metabolically engineered strains that are growth-coupled because production rates are directly related to growth rates of engineered strains.[26] In *Actinobacillus succinogenes*, it was determined that the step having the most significant positive effect on succinic acid production was glycerol uptake.[27] Another study carried out in *E. coli* using glycerol as a carbon source revealed that RNA polymerase mutants exhibited a 60% faster growth rate and converted their carbon source up to 15–35% more efficiently to biomass than their parent strain. However, this strategy also requires extensive laboratory time. The last strategy has the potential to reduce the cost and time required for experimentation because it utilizes computational metabolic models to predict a wide range of cellular functions and physicochemical (nutritional) requirements of microorganisms.[27–29] Metabolic reconstructions can be simulated using COnstraint-Based Reconstruction and Analysis (COBRA) methods. These simulations use mass constraints (other experimentally obtained, biologically related constraints can also be imposed on a model) and an objective function to generate metabolic flux distribution finally inside the cell under different scenarios (e.g., oxygen, magnesium, ammonia, and carbon limitation).[30]

One of the most commonly examined constraints imposed in metabolic models is the availability of substrate or nutrients in the environment. Environmental parameters may vary when performing COBRA simulations, computing growth states as a function of media composition. Metabolic modeling could be used to determine the nutritional needs of microbes with particular ecological and genomic traits by identifying a minimal set of reactions capable of supporting the minimal growth or by a gene-reaction association approach to identify the set of reactions needed to support the growth.[31,32] Nevertheless, they are designed only to predict viability.

Thus, further identification and evaluation of each compound available in the culture medium based on computational methods could serve as an attractive approach aiming to optimize growth conditions. Here, an iterative constraint matrix approach is proposed to evaluate extracellular compounds present in a culture medium using a metabolic model. This allows the identification of compounds that should be included in culture media to optimize the rate of glycerol uptake in *E. coli* as a microbial cell factory. This approach revealed that the glycerol uptake in *E. coli* is directly associated with the presence of nitrogen and phosphate, which has the potential to be linked to the generation of new, industry-attractive, bio-based compounds.

Materials and methods

Metabolic model refinement

A model, based on *EColiCore2* (ECC2),[33] was used. This model was updated based on the iML1515 model for biomass and incorporated for malate and formate exchange reactions.[34] However, Hädicke and Klamt reported that to avoid an undesired flux distribution of the model, which would result in an incorrect prediction, the network was developed without formate excretion and malate synthase.[33] The glyoxylate (glx[c]) and formate (for[c]) metabolites in glycolate oxidase (GLYCTO4), glyoxalate carboligase (GLXCL), 4-amino-2-methyl-5-phosphomethylpyrimidine synthetase (AMPMS2), 3,4-dihydroxy-2-butanone-4-phosphate synthase (DB4PS), formate-tetrahydrofolate ligase (FTHFLi), GTP cyclohydrolase I (GTPCI), and GTP cyclohydrolase II (25drapp) (GTPCII2) reactions were therefore modified by auxiliary metabolites glxBIO[c] and forBIO[c], respectively, as the authors suggested (supplementary file S1). For *E. coli* growing in glycerol, essential genes determined by the model, and the reactions associated with the genes reported as necessary by Joyce *et al.* were incorporated.[35]

In silico approach to determine the components of the media for the optimization of glycerol consumption

The optimization method shown in Fig. 1 was developed based on a phenotype phase plane analysis.[36] This identified the set of reactions that were directly related to the increases in glycerol uptake. The maximization of biomass as an objective function was quantified as a type of flux under a steady-state constrained mass balance similar to flux balance analysis (FBA). It involved two iterative fixed flux analyses that extended from 0 to the *nPts* flux boundary. The two iterative constraints used were fixed and included (i) the carbon source (glycerol) and (ii) one extracellular reaction in a selected subset of reactions, which represented each compound available in the culture medium. The optimization was expressed mathematically, as shown in Fig. 1. The method was implemented in MATLAB R2017b and was integrated into the COBRA Toolbox v.3.0.[36] using Gurobi 8.0.1 as a solver.

Flux balance analysis has been demonstrated to have a high capacity to predict cell behavior but it has some limitations mainly because it predicts the phenotype at a steady state.[37] Thus, the fluxes predicted are represented as either uptake or production rates. It cannot predict, in a dynamic state, metabolite concentrations available in the culture medium needed to support high glycerol consumption. Dynamic flux balance analysis (dFBA) is an extension of FBA that applies flux distribution constraints, allowing it to characterize dynamic concentration profiles and model a batch fermentative process via the incorporation of constraints,[8,38] and it involves the optimization of metabolites at a specific time. Once the medium components are determined, dFBA is used to predict the nutritional requirements in component concentrations that allow the cell to maximize glycerol consumption.

Strains and culture conditions for experimental validation

Escherichia coli ATCC 8739 was used in this study and was obtained from the American-type culture collection (ATCC). A glycerol-based medium was used containing 5 $g\,L^{-1}$ yeast extract, 2.5 $g\,L^{-1}$ NaCl, 5 $mL\,L^{-1}$ trace metals solution (0.55 $g\,L^{-1}$ $CaCl_2$, 0.10 $g\,L^{-1}$ $MnCl_2\ 4H_2O$, 0.17 $g\,L^{-1}$ $ZnCl_2$, 0.043 $g\,L^{-1}$ $CuCl_2\ 2H_2O$, 0.06 $g\,L^{-1}$ $CoCl_2\ 6H_2O$, 0.06 $g\,L^{-1}$ $Na_2MoO_4\ 2H_2O$, 0.06 $g\,L^{-1}$ $Fe(NH_4)_2(SO4)_2\ 6H_2O$, 0.20 $g\,L^{-1}$ $FeCl_3\ 6H_2O$), $MgSO_4$ 1 $mol\,L^{-1}$, 30 $g\,L^{-1}$ glycerol (Merck, USA). Some authors have reported that the supplementation of metals in low concentrations can stimulate growth while glycerol is used.[19] All conditions were supplemented with compounds at concentrations described in Table 1 for testing and validating the computational results based on a Taguchi experimental design. Aerobic cultures of 50 mL were carried out in 250 mL baffled-conical Erlenmeyer flasks at 37 °C and 200 rpm.

Analytical methods

One mL of culture was sampled at 12 and 24 h for cell estimation and high-performance liquid chromatography

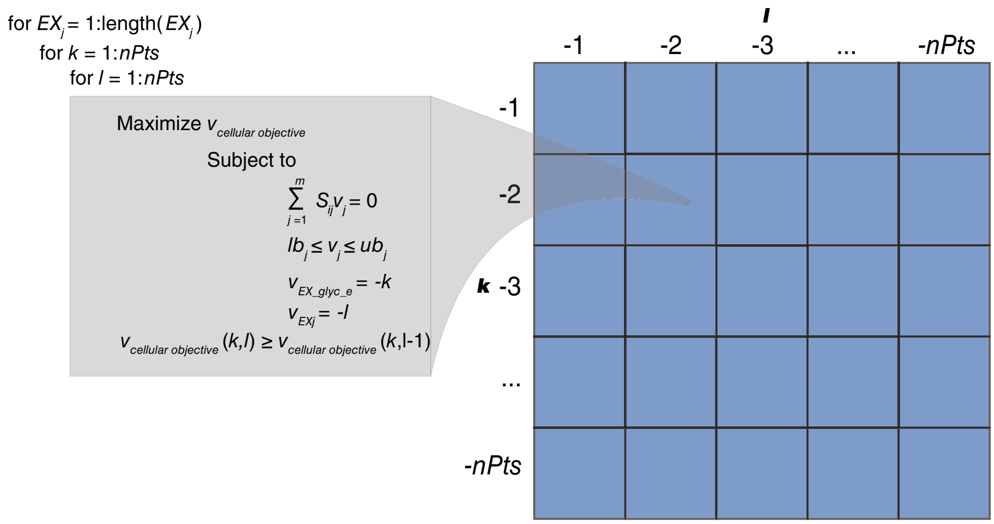


Figure 1. The framework used to evaluate extracellular compounds that increase levels of glycerol uptake has been depicted. *i* represents the metabolites and *j* indicates the reaction in the network. EX_j is the subset of extracellular reactions to be considered and *nPts* is the number of points and the flux boundary constraint to fix the flux for *EX_glyc_e* and each extracellular EX_j in each loop. $V_{cellular\ objective}$ is the objective function. *S* is a matrix that describes the metabolism of microorganisms considered. The size of the *S* matrix is *m*-by-*n*. stoichiometric coefficient, *m*, represents the number of metabolites, and *n* is the number of reactions in the model. S_{ij} is the stoichiometric coefficient of metabolite *i* in reaction *j*. v_j represents the flux of reaction *j*. The network is assumed to be at a quasi-steady state represent by $S_{ij}v_j = 0$. The lower and upper bounds of the flux of these reactions is described by lb_j and ub_j, respectively, which define the thermodynamic feasibility (directionality) and flux capacity of reaction *j*. Negative numbers represent maximum uptake and positive numbers represent secretion rates. $V_{cellular\ objective}^{(k,l)}$ is the biomass flux of the current nPts loop and $V_{cellular\ objective}^{(k,l-1)}$ is the flux of the previous loop.

(HPLC) analysis. Cell mass was estimated from optical density at 600 nm (OD_{600} 1.0 ~ 0.33 g of dry cell weight L^{-1}) with a UV-visible Thermo Scientific TM GENESYSTM 10 series spectrophotometer. After spectrophotometry, each sample was centrifuged at 1100 g for 8 min and stored at −20 °C for HPLC analysis. Then, culture supernatants were passed through a syringe filter (0.22 μm pore size) before HPLC analysis. Organic acids (succinic acid, acetic acid, lactic acid, fumaric acid, malic acid, pyruvic acid) and glycerol in extracellular samples were determined via HPLC using the Agilent 1260 Infinity system (Agilent Technologies, Santa Clara, California, USA). The system was equipped with an RI detector, operated at 45 °C and a 300 × 7.88 mm Aminex HPX-87H ion-exchange column (Bio-Rad Laboratories, USA). The column was operated at 30 °C with a flow rate of 0.5 mL min^{-1} using 5 mM H_2SO_4 as the mobile phase.

Table 1. The experimental design used to experimentally validate predicted optimized reactions for enhancing glycerol uptake in *Escherichia coli* under aerobic conditions.

Condition	NH_4Cl g L^{-1}	Na_2HPO_4 g L^{-1}	KH_2PO_4 g L^{-1}	Tryptone
1	1.0	1.2	0.6	0.0
2	1.0	3.6	1.8	0.0
3	1.0	6.0	3.0	0.0
4	3.0	1.2	1.8	0.0
5	3.0	3.6	3.0	0.0
6	3.0	6.0	1.2	0.0
7	5.0	1.2	3.0	0.0
8	5.0	3.6	1.2	0.0
9	5.0	6.0	2.4	0.0
10	0.0	0.0	0.0	10.0
11	0.0	0.0	0.0	0.0

Differential expression analysis

RNA-Seq was carried to examine gene expression in conditions 3 and 11, and biological triplicates were considered (Table 1). To harvest the cells to purify RNA after 6 h of culture, samples were first treated with RNAprotect bacterial reagent (Cat No./ID: 76506), and enzymatic lysis and Proteinase K digestion of bacteria was carried out following the manufacturer's protocol. Then, the Qiagen RNeasy Mini kit (Cat No./ID: 74104) was used to obtain total

 | *Biofuels, Bioprod. Bioref.* (2021); DOI: 10.1002/bbb.2205

RNA for further analysis according to the manufacturer's protocol. Each sample was treated with DNase to remove the genomic DNA. Samples were sent to commercial RNA-Seq services for further sample processing and sequencing using Illumina HiSeq 2x150 bp sequencing, and rRNA depletion was carried out in library preparation (Genewiz, South Plainfield, NJ, USA).

Clean raw data was obtained by removing reads that contained adapters using Trimmomatic v 0.38.[39] The sequence RefSeq: NC_CP010468 was employed for mapping. RNA reads were mapped using bowtie2 software,[40] and featureCounts[41] from subread 1.6 was applied to count reads. SARTools (Statistical Analysis of RNA-Seq data Tools)[42] was used for statistical RNA-Seq analysis. Differentially expressed genes (DEGs) were identified using the DESeq2[43] R package. The functional classification of the DEGs was performed using a cluster of orthologous groups (COG) analysis. The data discussed in this publication have been deposited in the (National Center for Biotechnology Information) NCBI's Gene Expression Omnibus[44] and are accessible through GEO Series accession number GSE140847.

Results

Metabolic network

After refinement process ECC2, a core model of *E. coli* was obtained, where all essential genes (72) reported experimentally for growth in glycerol were incorporated. It was found that the alteration of metabolites glx[c] (glyoxylate) and for[c] (formate) in GLYCTO4, GLXCL, AMPMS2, DB4PS, FTHFLi, GTPCI, and GTPCII2 reactions could generate new steady-state levels within flux distributions but could not in extracellular reactions. This new ECC2 model contained 593 metabolites and 655 reactions, where 211 were reversible, and 52 were extracellular reactions.

Some key properties of the network, such as reaction essentiality and metabolite production, were analyzed using FBA. Metabolic modeling using COBRA methods allows gene knockouts to be predicted by constraining each reaction's flux (based on the gene-protein-reaction relationship) to zero. Consequently, if the FBA solution or growth rate is unfeasible, the reaction is determined to be essential. Changes in the number of reactions or even reaction composition of the network may result in changes in solution space and flux distribution changes. Typically, efforts to identify essential reaction components have been focused on growth using glucose as a substrate. Glucose was the energy source for the initial ECC2 report.[33] As this study's objective was to enhance glycerol consumption and increase the networks' capacity to predict phenotypic states when glycerol was used as an energy source, 100 reactions associated with 72 genes reported to be essential for growth in glycerol by Joyce *et al.*[35] were incorporated. These included reactions that bridged disconnected metabolites. GapFind within the CobraToolbox was used to determine network gaps created by the incorporation of new reactions. GapFind is an optimization program that searches for root metabolites that are not connected to the network. Twelve metabolites (bis-molybdopterin guanine dinucleotide, bmocogdp[c]; cellobiose, cellb[e]; maltose 6′-phosphate, malt6p[c]; 2(alpha-D-mannosyl-6-phosphate)- D-glycerate, man6pglyc[c]; reduced thioredoxin, trdrd[c]; oxidized thioredoxin, trdox[c]; sucrose 6-phosphate, suc6p[c]; L-allo-threonine, athr__L[c]; *N*-acetylmuramate, acmum[e]; arbutin, arbt[e]; and maltose, malt[e]) were found to be disconnected, but they did not affect glycerol utilization.

After incorporating the reactions associated with essential genes tested experimentally, it was noted that associated changes in the network did not generate new flux distributions. Several COBRA-based methods, such as FBA, assume steady states and impose upper and lower bounds on metabolic fluxes; however, this approach typically does not consider thermodynamic restrictions explicitly, which can result in thermodynamically infeasible solutions. To avoid this issue, thermodynamically infeasible cycles (TICs) were identified. For all reactions in ECC2, minimal and maximal flux values were calculated using flux variability analysis (FVA).[45] For FVA, EX_glyc_e (glycerol) flux was fixed to -13.3 mmol/gDW-h^{-1} and oxygen was set to -1000 mmol/gDW-h^{-1} for aerobic conditions and 0 mmol/gDW-h^{-1} for anaerobic conditions. No reaction was observed reaching minimal (-1000) or/and maximal flux bounds (1000) in ECC2, indicating that no reactions were under TICs. Finally, to evaluate metabolite flux, an FBA was carried out using the COBRA Toolbox v.3.0.[36] Figure 2 shows the flux results of FBA for extracellular reactions under both aerobic and anaerobic conditions. The main by-products identified under anaerobic conditions were ethanol (6.49 mmol/gDW-h^{-1}), acetate (5.04 mmol/gDW-h^{-1}) and succinate (0.03 mmol/gDW-h^{-1}). Under aerobic conditions, no extracellular reactions associated with either organic acids or ethanol had a flux >0 mmol/gDW-h^{-1}.

Predicting cell requirements to optimize glycerol uptake

Extracellular reactions in ECC2 were used to determine the biomass requirements of *E. coli*. Nineteen reactions (EX_ca2_e, EX_cl_e, EX_co2_e, EX_cobalt2_e, EX_cu2_e, EX_fe2_e, EX_fe3_e, EX_h_e, EX_h2o_e, EX_k_e, EX_mg2_e, EX_mn2_e, EX_mobd_e, EX_nh4_e, EX_ni2_e,

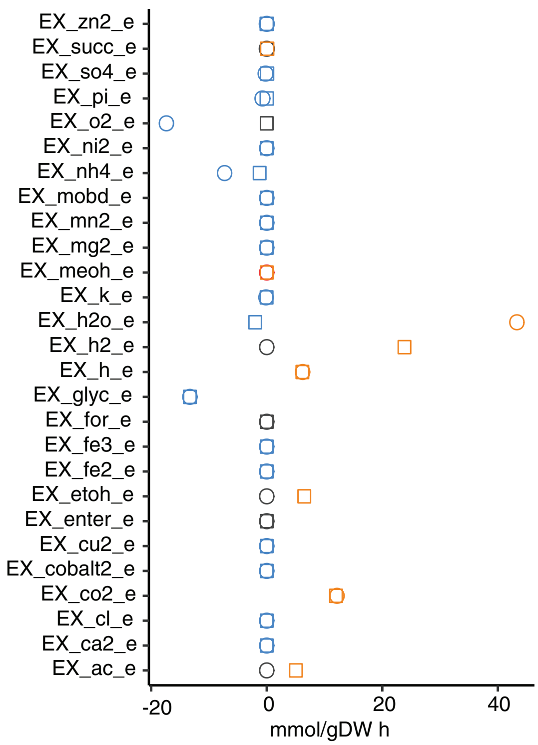


Figure 2. Flux balance analysis of ECC2 after refinement. Circles show fluxes under aerobic conditions and squares indicate fluxes under anaerobic conditions. Blue represents reactions / metabolites required, red shows reactions/ metabolites generated, and black values are equal to zero.

EX_pi_e, EX_so4_e, EX_zn2_e, EX_slnt_e) without carbon sources and oxygen sources were identified to be essential, as the determination of an optimal solution was unfeasible and associated fluxes were set between −1000 to 1000 using the minimal set reactions approach. These reactions were then used as subsets of reaction EX_j (Fig. 1). The lower bound for oxygen (EX_o2_e) was set to −20.00 mmol/gDW-h^{-1} in the model.

Figure 3 shows a phenotype phase plane (PhPP) for the growth rate of *E. coli*, glycerol uptake rate, and the uptake rate for each reaction predicted to have a higher effect on glycerol consumption. Three reactions (EX_nh4_e, EX_pi_e, EX_o2_e) were identified to have a high impact on both the rate of growth and glycerol uptake in ECC2. The greatest nutritional requirements needed to achieve maximal growth and glycerol uptake rates were observed when EX_o2_e was evaluated. The reaction required 20.00 mmol/gDW-h^{-1} of oxygen to reach an uptake rate of 20.00 mmol/gDW-h^{-1} of glycerol and 0.88/h growth rate (Fig. 3(G)). Oxygen was also observed to increase the glycerol uptake rate, together with biomass. EX_nh4_e was also seen significantly impacting the glycerol uptake rate as it was necessary to reach the maximum growth (0.88 1/h) and glycerol uptake (20.00 mmol/gDW-h^{-1}) rate, 9.57 mmol/gDW-h^{-1} . Figure 3(A) shows an increase with glycerol substrate and nitrogen uptake rates of up to 17.00 mmol/gDW-h^{-1} and 8.88 mmol/gDW-h^{-1}, respectively. After that, the nitrogen uptake rate increased further, up to 9.57 mmol/gDW-h^{-1}. In contrast to the EX_o2_e and EX_nh4_e, the phosphorous requirements (EX_pi_e) necessary to achieve maximal growth and glycerol uptake rates were low. In fact, 0.85 mmol/gDW-h^{-1} phosphorus was required at a 20.00 mmol/gDW-h^{-1} glycerol uptake rate (Fig. 3(D)).

To determine the effect of the addition of a given metabolite on the growth rate, shadow prices,[46] and the line of optimality were analyzed for the set of reactions EX_j reactions predicted to have a significant impact on glycerol uptake rates. Shadow prices are defined according to Eqn (1):

$$\pi_i = \frac{\partial \text{cellular objective}}{\partial v_i} \tag{1}$$

The line of optimality is defined as a line representing the optimal relationship between two metabolic fluxes (glycerol and extracellular reactions). It serves as the basis for identifying an optimal condition in a phenotype phase plane. Corresponding to the optimal level of conversion of glycerol into biomass, which is not accompanied by the formation of by-products.[47]

Figure 3(B),(C),(E),(F), (H), and (I) exhibit the shadow prices when EX_glyc_e, EX_nh4_e, EX_pi_e, and EX_o2_e are incrementally altered. There are five distinct phenotypes in the PhPP for glycerol and nitrogen uptake (Fig. 3(B)). The P1 observed in Fig. 3(B) and the nitrogen effect on the shadow prices in the same phase (Fig. 3(C)), highlights the importance of nitrogen for glycerol utilization since glycerol uptake directly increases the objective function. When comparing EX_glyc_e and EX_pi_e, one phase was observed for glycerol, and two phases for phosphorous shadow prices, showing that phosphorous has an increasing effect over the objective function in big jumps using glycerol as a carbon source. Five phases were exhibited when EX_glyc_e and EX_o2_e were incrementally constrained. The highest shadow prices were observed in P1–P3 for EX_o2_e versus EX_glyc_e. For P4, similar glycerol (0.021) and oxygen (0.025) shadow prices were obtained.

A robustness analysis was carried out to find a convergence point where growth and glycerol uptake rates were maximized, evaluating how nitrogen, phosphorous, and oxygen availability within the culture medium affect the glycerol uptake and growth rates. In this analysis, the flux through one reaction is varied, and the optimal growth rate is calculated as a function of this flux. Figure 3(J) shows a robustness analysis for EX_glyc_e, EX_nh4_e, EX_pi_e, and

 | *Biofuels, Bioprod. Bioref.* (2021); DOI: 10.1002/bbb.2205

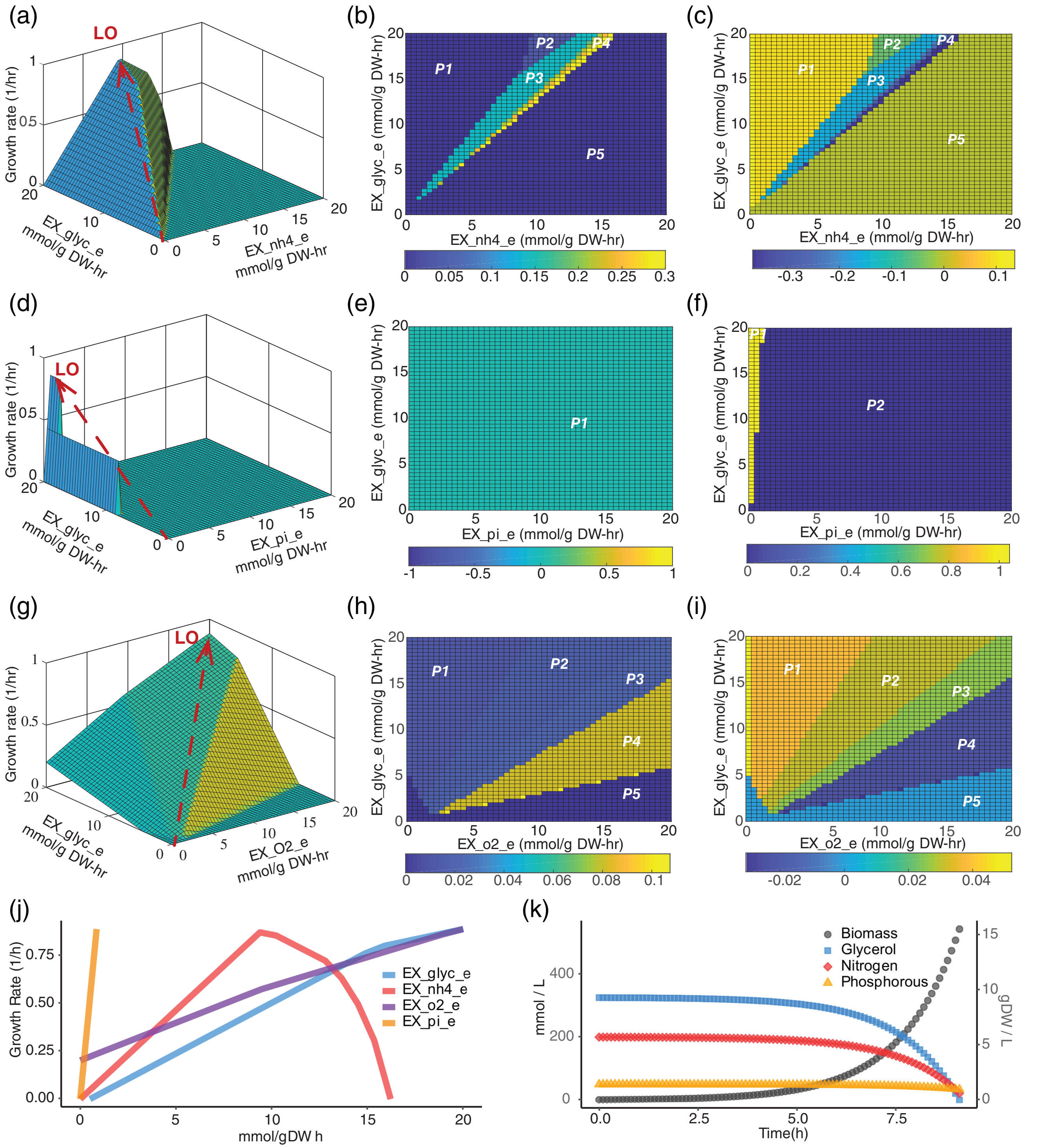


Figure 3. Phenotype plane phase, robustness, and dynamic flux balance of metabolites predicted to have a significant effect on glycerol uptake. Flux phenotype plane phases for the three most significant extracellular reactions predicted. EX_nh4_e, EX_pi_e, and EX_o2_e (A, D, and G, respectively) were compared by examining glycerol uptake and growth rates on the response surface. The red line represents the line of optimality. B, E, and H panels depict EX_glyc_e shadow prices for the three most significant extracellular reactions predicted, EX_nh4_e, EX_pi_e, and EX_o2_e, respectively. C, F, and I correspond to EX_nh4_e, EX_pi_e, and EX_o2_e shadow prices, respectively. J depicts an EX_nh4_e, EX_pi_e, and EX_o2_e robustness analysis and K includes dynamic flux balance analyses for EX_nh4_e and EX_pi_e reactions.

EX_o2_e. A continuous increasing tendency was observed for phosphorous and oxygen but the highest growth was exhibited for phosphorous. This result shows the same behavior as PhPP analysis and shadow prices. In contrast, as nitrogen uptake increases over 9.57 mmol/gDW-h^{-1}, non-biomass production or non-optimal biomass production is observed.

The dFBA was performed in MATLAB R2017b with the COBRA Toolbox v.3.0[36] and Gurobi 8.0.1 solver. The method was comprised of 325 mmol glycerol (EX_glyc_e), 200 mmol nitrogen (EX_nh4_e), and 50 mmol trace elements, including phosphorous (EX_ca2_e, EX_cl_e, EX_cobalt2_e, EX_cu2_e, EX_fe2_e, EX_fe3_e, EX_k_e, EX_mg2_e, EX_mn2_e, EX_mobd_e, EX_ni2_e, EX_pi_e, EX_so4_e, EX_zn2_e, EX_slnt_e), which were used at initial concentrations in the culture medium, and uptake rates were defined using the same methods that were used to perform the FBA. Figure 3(K) shows the results obtained in the dFBA. The dFBA suggested that, to achieve a consumption of 325 mmol at one uptake rate of 13.3 mmol/gDW-h^{-1} of glycerol, 189 mmol of nitrogen, and 16 mmol of the phosphorous source would be required. The other essential compounds evaluated seem to be necessary – however, at minimal concentrations of 1 mmol.

Experimental determination of the effects of nitrogen, and phosphorous over glycerol uptake

Computational modeling departing from a PhPP analysis suggested that nitrogen, oxygen, and phosphorus play a significant role regarding nutritional requirements that allow *E. coli* to take up glycerol under aerobic conditions optimally. NH_4Cl, Na_2HPO_4, and K_2HPO_4 were nutrients selected to determine the nitrogen and phosphorous effects over the growth rate experimentally. NH_4Cl was added to culture media to act as a nitrogen source and was denoted as the EX_nh4_e reaction. Na_2HPO_4, and K_2HPO_4 were employed as a phosphorus source, denoted in the metabolic network as EX_pi_e,. Finally, for EX_o2_e, a baffled-conical flask and 200 rpm were used to supply the necessary oxygen. Previous studies have shown that baffled conical flasks can meet the oxygen demands of *E. coli*.[48] The highest concentration shown in Table 1 were based on those reported using M9 medium[49] and covered those results obtained by the dFBA. The glycerol-based culture medium was supplemented with low levels of trace metals and $MgSO_4$, which were predicted to be essential for *E. coli* growth, and dFBA suggested they were needed at low concentration.

For the glycerol-based medium design, 33 cultures were tested (three biological replicates for each condition described in Table 1), and the profiles obtained for glycerol consumption are shown in Fig. 4. For conditions 1, 2, 4, 5, 7, 8, 10, and 11 the glycerol uptake rate was 9.95 mmol/gDW-h^{-1} consuming 11.00 ± 1.29 g L^{-1} at glycerol at an uptake rate of mmol/gDW-h^{-1} after 12 h of culture and reaching stationary phase (Fig. 4(B)). Although similar glycerol consumption levels were observed after 12 h in condition 11, *E. coli* continued consuming glycerol for up to 24 h (Fig. 4(B),(E)). Accordingly with the model simulation no production of by-products was detected in these conditions, as it predicted that no acetate generation when glycerol uptake rates were under 16 mmol/gDW-h^{-1}.

Na_2HPO_4 and KH_2PO_4 play essential roles in glycerol uptake. Figure 4(A) shows a cluster analysis, including NH_4Cl, Na_2HPO_4, KH_2PO_4, tryptone, and glycerol consumption variables. This result indicates that phosphorus sources in the form of either Na_2HPO_4 or KH_2PO_4 appear to be required at significantly increased levels for glycerol consumption, accordingly with the fact that EX_pi_e increases as biomass increases (Fig. 3(J)). The maximum growth and glycerol consumption rates were obtained in culture media (conditions 3, 6, and 9) containing the maximum concentration of Na_2HPO_4 (Fig. 4(B), Table 1). Nevertheless, the highest glycerol consumption levels were achieved for the condition with the maximum availability of KH_2PO_4 (Table 1). The glycerol uptake rate for condition 3 was 23.49 mmol/gDW-h^{-1} and was approximately 18.09 mmol/gDW-h^{-1} for conditions 6 and 9. Interestingly, the culture medium used in condition 3 was promising, and *E. coli* consumed up to 29.70 g L^{-1} after 24 h of growth (Fig. 4(C)). However, the generation of acetate was predicted by computation results. Comparing condition 3 (Fig. 4(C)) with LB medium, a standard growth medium for microorganisms, supplemented with trace metals and $MgSO_4$ (condition 10) (Fig. 4(D)), a twofold improvement in glycerol consumption was observed.

RNA-Seq and cluster of orthologous groups analysis

To understand and validate our hypothesis that the amount of nitrogen, phosphorous, and oxygen are positively related to glycerol consumption, determining the expression of critical genes associated with the increased glycerol uptake (which achieved a glycerol uptake rate of 23.49 mmol/gDW-h^{-1} in condition 3), a transcriptomic analysis was carried out. The presence of DEGs was determined using the deseq2 statistical package using condition 11 as a reference after filtering out

  | *Biofuels, Bioprod. Bioref.* (2021); DOI: 10.1002/bbb.2205

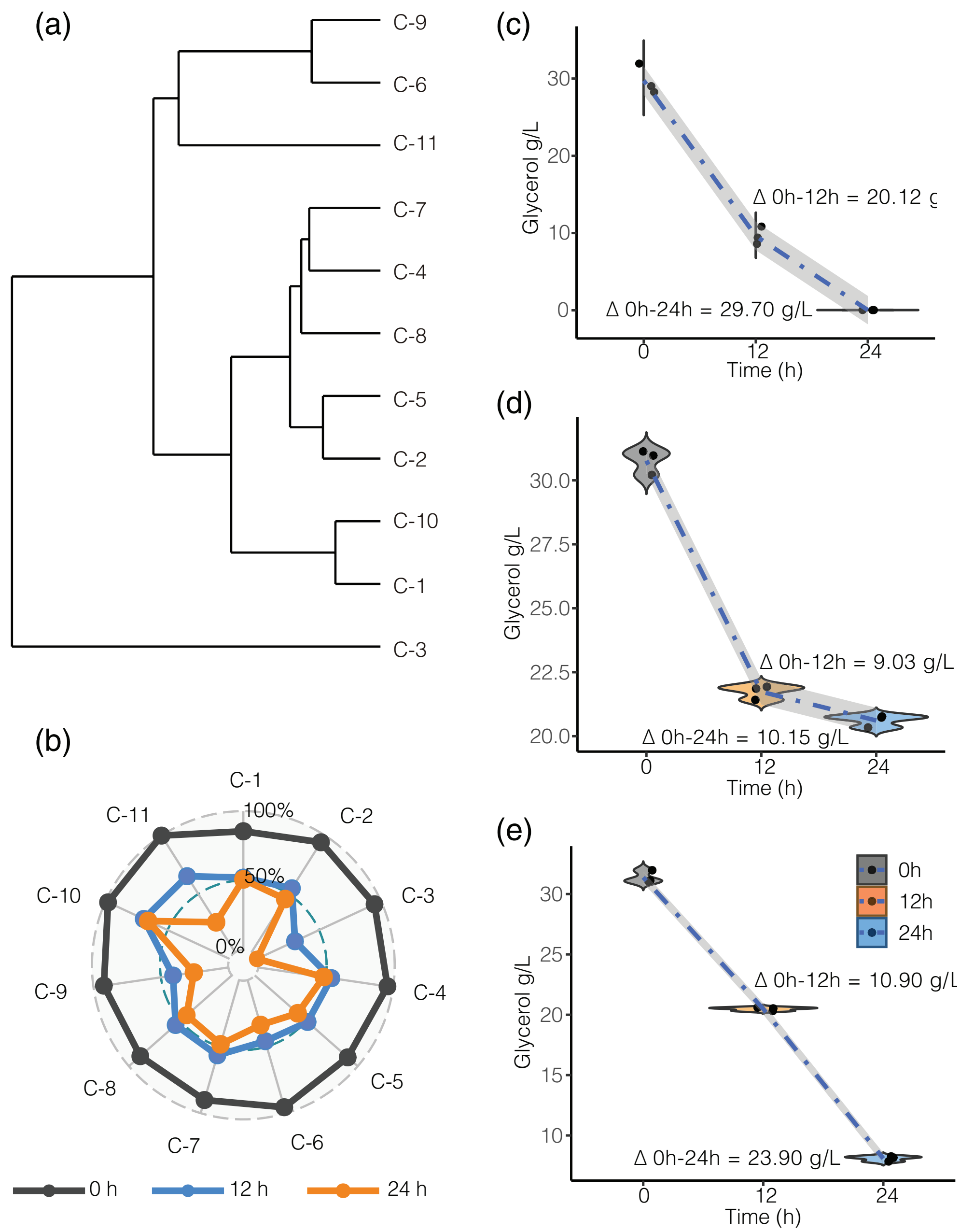


Figure 4. Experimental glycerol consumption by *Escherichia coli* ATCC 8739 under aerobic conditions based on optimization method predictions. (A) Cluster analysis for the conditions listed in Table 1, which examined use of NH_4Cl, Na_2HPO_4, KH_2PO_4, tryptone, and glycerol consumed variables. (B) Percentage of glycerol consumption for the conditions listed in Table 1. All the data were normalized to a 35 g L^{-1} initial glycerol value. (C) Glycerol consumed throughout condition 3. (D) Glycerol consumed throughout condition 10. (E) Glycerol consumed throughout condition 11.

low-count reads with an average value of ≤100. Figure 5(A) shows DEG levels using a Log_2 FC ≥ |2| scale. This analysis determined that 478 genes were differentially expressed, with 222 genes down-regulated and 256 up-regulated.

Diferentially expressed genes were clustered using the COG approach (Fig. 5(B)). This analysis revealed that 41.5% of down-regulated genes had been poorly characterized. Further, 79% of those genes had not been mapped to a COG category, probably because they were categorized as hypothetical proteins, and 38.7% of down-regulated genes were grouped classified as metabolic genes, and 12.6% were predicted to be involved in cellular processes and signaling. The remaining genes were grouped classified as information storage and processing genes.

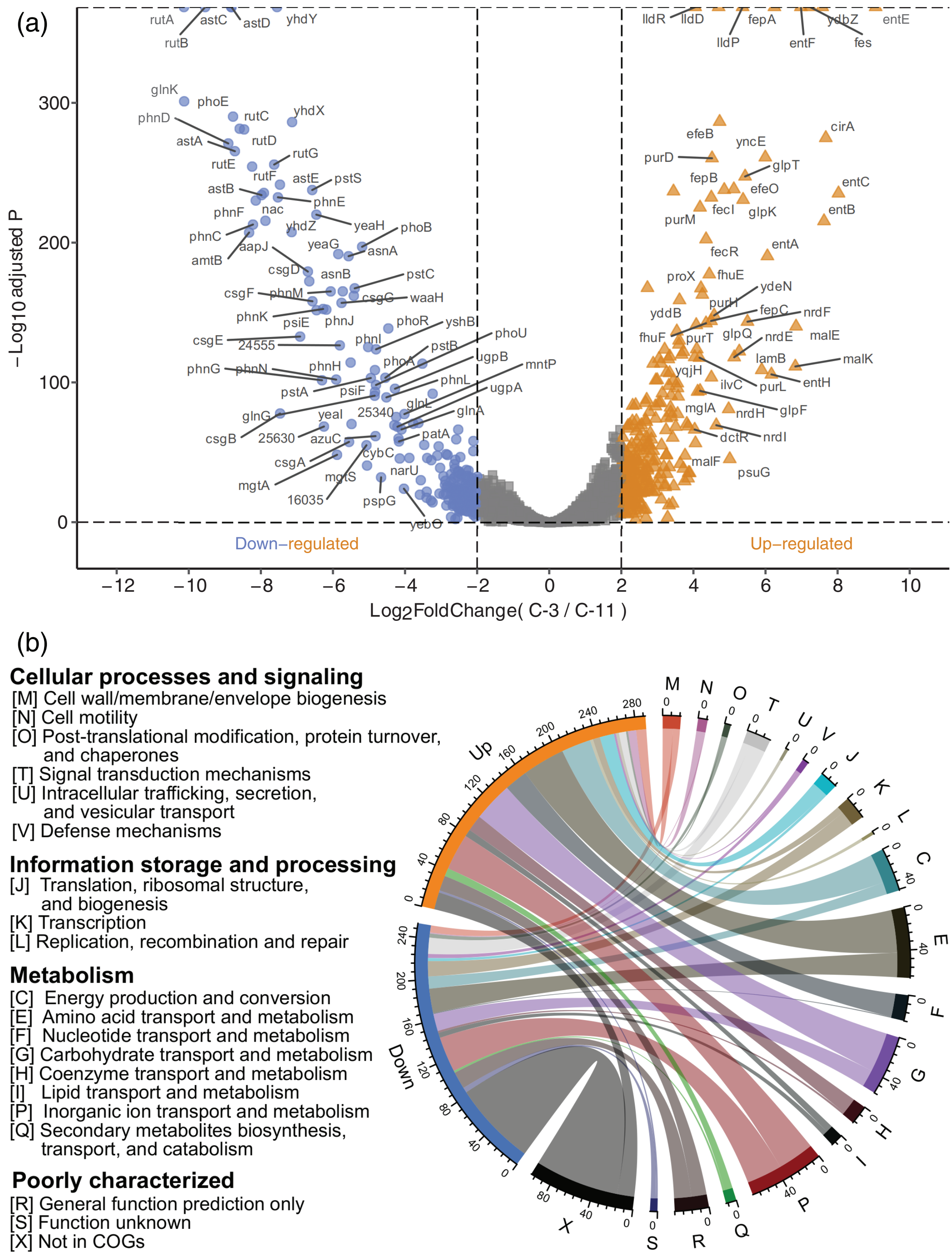


Figure 5. DEGs obtained from comparisons made between condition 3 and condition 11. (A) Volcano plot of DEGs in which down-regulated genes are shown in blue, and up-regulated genes are shown in red. (B) Cluster of orthologous groups of DEGs.

In contrast, the most considerable portion of up-regulated genes was classified as cell metabolism genes (69%). Further, 11% and 8.3% were predicted to be genes involved in cellular processes and signaling and information storage and processing, respectively. Finally, 11.7% of up-regulated genes identified had been poorly characterized.

Biofuels, Bioprod. Bioref. (2021); DOI: 10.1002/bbb.2205

Discussion

To obtain energy and synthesize cellular molecules, microorganisms must be supplied with nutrients. The importance of any culture medium containing a carbon source is well known as all cells need large amounts of carbon to generate new cellular materials.[18,50] However, oxygen, hydrogen, nitrogen, sulfur, phosphorus, potassium, calcium, magnesium, and iron are also needed, along with micronutrients such as manganese, zinc, cobalt, molybdenum, nickel, and copper. Overall, an increase in growth and glycerol uptake rates was observed when nitrogen, phosphorus, and oxygen uptake rates increased concomitantly. Although reactions EX_ca2_e, EX_cl_e, EX_cobalt2_e, EX_cu2_e, EX_fe2_e, EX_fe3_e, EX_k_e, EX_mg2_e, EX_mn2_e, EX_mobd_e, EX_ni2_e, EX_so4_e, EX_zn2_e, EX_slnt_e were identified as essential for *E. coli* growth on glycerol, they were not determined to have a critical impact on improving glycerol uptake. This is likely because most are micronutrients, such as manganese, zinc, cobalt, molybdenum, nickel, and copper, which are essential for the growth of *E. coli* but are not associated with increases in the glycerol uptake.

Shadow price and robustness analysis suggest that the biomass limitation in Fig. 3(B) is due to the nitrogen uptake rate because, at about 9.00 mmol/gDW-h^{-1}, the optimal line is reached. Increases in the nitrogen uptake rates after that limit appear to be a limiting factor concerning biomass generation from glycerol in *E. coli*. Moreover, it increases the nitrogen uptake rate limit of 8.50 mmol/gDW-h^{-1} and results in acetate production under aerobic conditions. This could become a potential bottleneck for producing of other organic acids from glycerol because it could affect bioconversion yields, and the recovery and purification process. Moreover, positive shadow prices observed in In P2 and P4 suggests that glycerol becomes inhibitory, probably because oxygen is limited (the maximum oxygen uptake rate for all the simulations was up to −20.00 mmol/gDW-h^{-1}). These phases are expected to be phenotypically unstable.[47] Thus, glycerol and nitrogen consumption are not needed to improve the objective function.[47] As a result, nitrogen may be a crucial compound taken up in *E. coli* cultures using glycerol as a carbon source.

The highest shadow prices were observed for oxygen, meaning that increasing oxygen levels significantly affected more biomass production, compared to nitrogen and phosphorous due to negative values obtained in the ratio of relative shadow prices,[47] despite dual substrate limitation. This phenomenon is plausible because oxygen acts as a final electron acceptor as glycerol is utilized.[12] However, the growth in phase 4 was limited by excess oxygen, even though the difference between shadow prices in phenotypic phase 4 was in favor of oxygen, and it remains the most valuable component. In this case, there were not enough glycerol molecules available to reduce available oxygen and produce biomass optimally.

Finally, although phosphorous is required as a micronutrient, the results indicate that small changes in the phosphorous uptake rates will rapidly increase biomass production and glycerol uptake rates. This has been suggested as essential to avoid cell toxicity during glycerol metabolism because of the generation of glycerol-3-phosphate (Gly-3-P), which could explain the *in silico* results obtained here.[51] Interestingly, high cell density cultures were observed, experimentally, which could be strongly related to the presence of phosphorus because it has been reported that to obtain denser cultures, phosphorus, sulfur, and trace elements must be added.[50] These are similar to the *in silico* results obtained here (Fig. 3(D)).

The use of some metabolites was predicted to become inhibitory to *E. coli* at high concentrations. Flux balance analysis and dFBA have a strong capacity to predict the behavior of reactions as a one-level system inside the cell.[52,53] However, they are limited by their failure to consider regulatory and inhibitory phenomena that occur within the cell. An experimental design was used to test and compare the results predicted, determining the concentration of nutrients required, while simultaneously avoiding toxic effects. As a result, the inclusion of a nitrogen source in the media is crucial for growth (Fig. 3(A)–(C)). For *E. coli*, ammonia is the nitrogen source that preferentially supports the microorganism's most rapid growth. Experimental observations regarding bacterial growth have suggested that *E. coli* maintains optimal growth under a wide range of external ammonia concentrations.[54] In this study, it was predicted that nitrogen uptake rates, reaching more than 9.57 mmol/gDW-h^{-1}, negatively impact growth rates, which will decrease to zero (Fig. 3(J)). The same effect was observed for experimental results, because the lowest concentration of nitrogen was required for condition 3, suggesting that supplementation of NH_4CL may harm glycerol consumption but not biomass formation. High generation of acetate was also observed in the experimental results decreasing the pH, similar to the computational modeling obtained when reaching more than 9.57 mmol/gDW-h^{-1}. Low pH produces one of the bottlenecks associated with bio-based chemical production because it reduces biomass formation. Thus, the addition of a phosphate buffer system using the salts Na_2HPO_4 and KH_2PO_4 could provide environmental conditions that are favorable to biomass formation with a buffering capacity similar to 89 mM phosphate.[55] In this study, phosphate supplementation in the condition 3 culture medium for glycerol consumption was 66 mM. The presence of a high-capacity phosphate buffer system could reduce the toxicity of the acidic metabolites secreted, which could be advantageous for organic acid production by using *E. coli* as a microbial cell factory. Dynamic flux balance analysis suggested that, for the consumption of 30 g of glycerol, 189 mmol of nitrogen was required. However, experimental results indicated that 55 mmol of NH_4 was needed. These discrepancies are likely due to yeast extract, a complex nitrogen source, which provides nitrogen in the form of amino acids and facilitates the rapid growth of microorganisms to grow faster, and provides nitrogen requirements.[56] Additional supplementation of nitrogen in the form of NH_4Cl would not have had as significant an impact on glycerol uptake when media was supplemented with yeast extract.[55]

Glycerol utilization of approximately 10 g L^{-1} using a medium containing high yeast extract concentrations and tryptone was observed.[12] However, when media supplemented with yeast extract (Fig. 4(D)) are compared with media supplemented with yeast extract and tryptone (Fig. 4(E)), tryptone appears to hinder glycerol uptake (Fig. 4(B)) (23.29 versus 10.15 g L^{-1}, respectively). These results suggest that using a restricted medium would make glycerol utilization as a carbon source for the generation of bio-based novel chemical products more amenable to industrial processes, especially regarding downstream processing steps.

It was hypothesized that the computational results have a high-level of predictive value because, in the metabolism of *E. coli*, oxygen acts as a final electron acceptor when glycerol is utilized as a carbon source.[12] This oxygen should be converted to water via the conversion of NADH to NAD+. Subsequently, oxygen appears to be strictly related to increasing glycerol uptake rate, as shown in Fig. 3(G). GlpF facilitates glycerol uptake in *E. coli*. Next, glycerol is transformed into Gly-3-P by GlpK under aerobic conditions, which is the first intermediate in the pathway produced before entry into the TCA cycle. It is also a point of convergence with both the biosynthesis and catabolism of lipids.[51] However, the accumulation of Gly-3-P in the cytoplasm is carefully controlled to avoid toxicity.[51] Externally generated Gly-3-P is then transported by GlpT, which exchanges a phosphate to prevent Gly-3-P and inorganic phosphate toxicity. This could explain why phosphorus is an essential biochemical (nutritional) requirement, enhancing glycerol uptake in *E. coli*. This is in addition to the requirement for NADP/NADPH in ammonia assimilation and biomass generation. Once Gly-3-P is in the cytoplasm, it is oxidized by GlpD in the presence of either oxygen or nitrates, resulting in the transfer of electrons to respective terminal oxidizers.

Within the metabolism group, categories with the highest number of clustered, up-regulated genes were those associated with inorganic ion transport and metabolism, amino acid transport and metabolism, carbohydrate carbon and metabolism, and energy production and conversion, these being most of the up-regulated genes. This result confirms the use of *glpF* (Log_2 FC 4.17) in *E. coli* for glycerol transport (Fig. 5(A)). Once glycerol is in the cell, it is phosphorylated to Gly-3-P by *glpK* (Log_2 FC 5.36) for further use in glycolysis as well as biosynthesis and catabolism of lipids. High levels of expression of the *glpT* (Log_2 FC 5.41) gene also result in the aerobic expression of GlpT, an sn-glycerol 3-phosphate transport enzyme induced by the generation of Gly-3-P substrate by *glpQ*.[57,58] This phenomenon could explain the phosphorus requirement for high glycerol uptake rates as glpT activity requires phosphorus for gradient transport.

Nitrogen is an essential element for all microorganisms. It is required for rapid growth. In *E. coli*, carbon and nitrogen metabolism have shared regulatory components.[59] This shared regulatory strategy involves a critical branch point between carbon and nitrogen utilization pathways in which α-ketoglutarate (αKG) serves both as a TCA intermediate and as the carbon backbone of glutamate and glutamine, the main nitrogen currencies.[59] In *E. coli*, isocitrate dehydrogenase (*icd*) is responsible for the generation of α-ketoglutarate. The *icd* gene achieves a Log_2 FC value of 1.47, and this showed that nitrogen was required for enhancing glycerol utilization. In the TCA cycle, s*ucA*BCD genes regulate *icd* activity to generate succinyl-CoA, a succinate precursor, which has received attention in bio-based industrial applications. The genes are up-regulated with Log_2 FC values of 2.13, 2.26, 2.69, and 2.59, respectively. This up-regulation of *icd* and s*ucA*BCD genes is an interesting finding that could result in a remarkable perspective on further metabolic engineering processes such as succinic acid production from glycerol. Finally, *E. coli* growth in a minimal medium is slower than in rich medium and other sugars.[60] In our study, to support fast growth without affecting glycerol uptake, the glycerol-based medium was supplemented with yeast extract. It is a complex nitrogen source, but yeast extracts also provide vitamins and amino acids. As a result, a significant number of genes in the COG category E were up-regulated. This suggest that supplementation of yeast extract supports the cell with essential amino acids needed to generate biomass faster and at low cost in any event that a cell prefers to import a substance rather than synthesize it.[61]

  | *Biofuels, Bioprod. Bioref.* (2021); DOI: 10.1002/bbb.2205

Conclusion

Optimization of even a few medium components is a challenging, time consuming, and costly task, because of the interactions between parts. Computational modeling is a great alternative to understand the cell and minimize this challenge. However, the use of this strategy has some limitations mainly associated with the regulation phenomena because a clear relationship exists between cell growth, product synthesis, and gene regulation. The use of a transcriptional response provides essential insights into the whole metabolism of *E. coli* when using glycerol as a carbon source. It allows an understanding of the way in which the transcriptomic level influences the metabolism in a cell and how it could help to formulate or reformulate potential culture media for industrial purposes.

In this study, the phenotype plane phase analyses allowed the development of a new potential approach to design and optimize culture media departing from the central metabolism. Although these analyses do not accurately determine the concentrations of compounds in the environment (culture media), they serve as a guide to identifying chemical compounds to be used in the laboratory for a media formulation, thus minimizing time, labor, and costs.

Acknowledgements

We are grateful to the Gobernación del Cesar, which supported this research under the PhD scholarship program 681 of the Colombian Ministry of Science, Technology and Innovation and the Instituto Amazónico de Investigaciones Científicas (SINCHI) for support with chromatographic analysis.

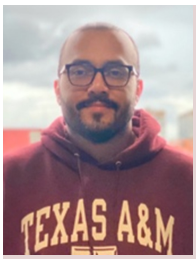
References

1. Cherubini F, The biorefinery concept: Using biomass instead of oil for producing energy and chemicals. *Energy Conver Manage* **51**(7):1412–1421 (2010).
2. Vlysidis A, Binns M, Webb C and Theodoropoulos C, A techno-economic analysis of biodiesel biorefineries: Assessment of integrated designs for the co-production of fuels and chemicals. *Energy* **36**(8):4671–4683 (2011).
3. Quispe CAG, Coronado CJR and Carvalho JA, Glycerol: Production, consumption, prices, characterization and new trends in combustion. *Renew Sustain Energy Rev* **27**:475–493 (2013).
4. Zhang X, Shanmugam KT and Ingram LO, Fermentation of glycerol to succinate by metabolically engineered strains of *Escherichia coli*. *Appl Environ Microbiol* **76**(8):2397–2401 (2010).
5. Ayoub M and Abdullah AZ, Critical review on the current scenario and significance of crude glycerol resulting from biodiesel industry towards more sustainable renewable energy industry. *Renew Sustain Energy Rev* **16**(5):2671–2686 (2012).
6. Ciriminna R, Pina CD, Rossi M and Pagliaro M, Understanding the glycerol market. *Eur J Lipid Sci Technol* **116**(10):1432–1439 (2014).
7. Serrano-Bermúdez LM, González Barrios AF, Maranas CD and Montoya D, Clostridium butyricum maximizes growth while minimizing enzyme usage and ATP production: Metabolic flux distribution of a strain cultured in glycerol. *BMC Syst Biol* **11**(1):58 (2017).
8. Tafur Rangel AE, Camelo Valera LC, Gómez Ramírez JM and González Barrios AF, Effects of metabolic engineering on downstream processing operational cost and energy consumption: The case of *Escherichia coli's* glycerol conversion to succinic acid. *J Chem Technol Biotechnol* **93**(7):2011–2020 (2018).
9. Tan HW, Abdul Aziz AR, Aroua MK, Aziz ARA, Aroua MK, Abdul Aziz AR *et al.*, Glycerol production and its applications as a raw material: A review. *Renew Sustain Energy Rev* **27**:118–127 (2013).
10. Chaudhary N, Ngadi MO and Simpson B, Comparison of glucose, glycerol and crude glycerol fermentation by *Escherichia coli* K12. *J Bioprocess Biotechnol* **s1**(01):1–5 (2014).
11. Sauer M, Porro D, Mattanovich D and Branduardi P, Microbial production of organic acids: Expanding the markets. *Trends Biotechnol* **26**(2):100–108 (2008).
12. Dharmadi Y, Murarka A and Gonzalez R, Anaerobic fermentation of glycerol by *Escherichia coli*: A new platform for metabolic engineering. *Biotechnol Bioeng* **94**(5):821–829 (2006).
13. Yin X, Li J, Shin HD, Du G, Liu L and Chen J, Metabolic engineering in the biotechnological production of organic acids in the tricarboxylic acid cycle of microorganisms: Advances and prospects. *Biotechnol Adv* **33**(6):830–841 (2015).
14. Mienda BS, Shamsir MS and Md. Illias R, Model-aided atpE gene knockout strategy in *Escherichia coli* for enhanced succinic acid production from glycerol. *J Biomol Struct Dyn* **34**(8):1705–1716 (2016).
15. Kim NY, Kim SN and Kim OB, Long-term adaptation of *Escherichia coli* to methanogenic co-culture enhanced succinate production from crude glycerol. *J Ind Microbiol Biotechnol* **45**(1):71–76 (2018).
16. Conrad TM, Lewis NE and Palsson BO, Microbial laboratory evolution in the era of genome-scale science. *Mol Syst Biol* **7**(1):509–509 (2011).
17. Murarka A, Dharmadi Y, Yazdani SS and Gonzalez R, Fermentative utilization of glycerol by Escherichia coli and its implications for the production of fuels and chemicals. *Appl Environ Microbiol* **74**(4):1124–1135 (2008).
18. Thomas E, Microbial growth and physiology: A call for better craftsmanship. *Front Microbiol* **6**(MAR):287 (2015).
19. Trchounian K, Poladyan A and Trchounian A, Optimizing strategy for *Escherichia coli* growth and hydrogen production during glycerol fermentation in batch culture: Effects of some heavy metal ions and their mixtures. *Appl Energy* **177**:335–340 (2016).
20. Biniarz P, Coutte F, Gancel F and Łukaszewicz M, High-throughput optimization of medium components and culture conditions for the efficient production of a lipopeptide pseudofactin by *Pseudomonas fluorescens* BD5. *Microb Cell Fact* **17**(1):121 (2018).
21. Cofré O, Ramírez M, Gómez JM and Cantero D, Optimization of culture media for ethanol production from glycerol by *Escherichia coli*. *Biomass Bioenergy* **37**:275–281 (2012).

22. Dragosits M and Mattanovich D, Adaptive laboratory evolution - principles and applications for biotechnology. *Microb Cell Fact* **12**(1):64 (2013).
23. Reyes LH, Winkler J and Kao KC, Visualizing evolution in real-time method for strain engineering. *Front Microbiol* **3**(MAY):1–6 (2012).
24. Winkler J, Reyes LH and Kao KC, Adaptive laboratory evolution for strain engineering. *Methods Mol Biol* **985**:211–222 (2013).
25. Reyes LH and Kao KC, Growth-coupled carotenoids production using adaptive laboratory evolution, in *Methods in Molecular Biology*. Humana Press, New York, NY, pp. 319–330 (2018).
26. Portnoy VA, Bezdan D and Zengler K, Adaptive laboratory evolution-harnessing the power of biology for metabolic engineering. *Curr Opin Biotechnol* **22**(4):590–594 (2011).
27. Binns M, Vlysidis A, Webb C, Theodoropoulos C, de Atauri P and Cascante M, Glycerol metabolic conversion to succinic acid using Actinobacillus succinogenes a metabolic network-based analysis. *Comput Aided Chem Eng* **29**:1421–1425 (2011).
28. Mora Salguero DA, Fernández-Niño M, Serrano-Bermúdez LM, Páez Melo DO, Winck FV, Caldana C *et al*., Development of a *Chlamydomonas reinhardtii* metabolic network dynamic model to describe distinct phenotypes occurring at different CO2 levels. *PeerJ* **6**:e5528 (2018).
29. Serrano-Bermúdez LM, Barrios AFG and Montoya D, Clostridium butyricum population balance model: Predicting dynamic metabolic flux distributions using an objective function related to extracellular glycerol content. *PLoS One* **13**(12):e0209447 (2018).
30. Maia P, Rocha M and Rocha I, In silico constraint-based strain optimization methods: The quest for optimal cell factories. *Microbiol Mol Biol Rev* **80**(1):45–67 (2016).
31. Burgard AP, Vaidyaraman S and Maranas CD, Minimal reaction sets for Escherichia coli metabolism under different growth requirements and uptake environments. *Biotechnol Prog* **17**(5):791–797 (2001).
32. Zarecki R, Oberhardt MA, Reshef L, Gophna U and Ruppin E, A novel nutritional predictor links microbial fastidiousness with lowered ubiquity, growth rate, and cooperativeness. *PLoS Comput Biol* **10**(7):e1003726 (2014).
33. Hädicke O and Klamt S, EColiCore2: A reference network model of the central metabolism of Escherichia coli and relationships to its genome-scale parent model. *Sci Rep* **7**(November 2016):39647 (2017).
34. Monk JM, Lloyd CJ, Brunk E, Mih N, Sastry A, King Z *et al*., iML1515, a knowledgebase that computes *Escherichia col*i traits. *Nat Biotechnol* **35**(10):904–908 (2017).
35. Joyce AR, Reed JL, White A, Edwards R, Osterman A, Baba T *et al*., Experimental and computational assessment of conditionally essential genes in Escherichia coli. *J Bacteriol* **188**(23):8259–8271 (2006).
36. Heirendt L, Arreckx S, Pfau T, Mendoza SN, Richelle A, Heinken A *et al*., Creation and analysis of biochemical constraint-based models using the COBRA toolbox v.3.0. *Nat Protoc* **14**(3):639–702 (2019).
37. Segre D, Vitkup D and Church GM, Analysis of optimality in natural and perturbed metabolic networks. *Proc Natl Acad Sci U S A* **99**(23):15112–15117 (2002).
38. Mahadevan R, Edwards JS and Doyle FJ, Dynamic flux balance analysis of diauxic growth in *Escherichia coli*. *Biophys J* **83**(3):1331–1340 (2002).
39. Bolger AM, Lohse M and Usadel B, Trimmomatic: A flexible trimmer for Illumina sequence data. *Bioinformatics* **30**(15):2114–2120 (2014).
40. Langmead B and Salzberg SL, Fast gapped-read alignment with bowtie 2. *Nat Methods* **9**(4):357–359 (2013).
41. Liao Y, Smyth GK and Shi W, FeatureCounts: An efficient general purpose program for assigning sequence reads to genomic features. *Bioinformatics* **30**(7):923–930 (2014).
42. Varet H, Brillet-Guéguen L, Coppée JY and Dillies MA, SARTools: A DESeq2- and edgeR-based R pipeline for comprehensive differential analysis of RNA-Seq data. *PLoS One* **11**(6):e0157022 (2016).
43. Love MI, Huber W and Anders S, Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol* **15**(12):550 (2014).
44. Edgar R, Gene expression omnibus: NCBI gene expression and hybridization array data repository. *Nucleic Acids Res* **30**(1):207–210 (2002).
45. Gudmundsson S and Thiele I, Computationally efficient flux variability analysis. *BMC Bioinform* **11**(2):2–4 (2010).
46. Schellenberger J, Que R, Fleming RMT, Thiele I, Orth JD, Feist AM *et al*., Quantitative prediction of cellular metabolism with constraint-based models: The COBRA toolbox v2.0. *Nat Protoc* **6**(9):1290–1307 (2011).
47. Palsson BO, *Systems Biology Constraint-Based Reconstruction and Analysis*. Cambridge University Press, Cambridge (2015).
48. McDaniel LE and Bailey EG, Effect of shaking speed and type of closure on shake flask cultures. *Appl Microbiol* **17**(2):286–290 (1969).
49. Bernstein HC, Carlson RP. Engineering and analyzing multicellular systems. In pp. 49–68. (2014).
50. Shiloach J and Fass R, Growing *E. coli* to high cell density - a historical perspective on method development. *Biotechnol Adv* **23**(5):345–357 (2005).
51. Booth IR, Glycerol and methylglyoxal metabolism. *EcoSal Plus* **1**(2):1–8 (2014).
52. Campbell K, Xia J and Nielsen J, The impact of systems biology on bioprocessing. *Trends Biotechnol* **35**(12):1156–1168 (2017).
53. Flassig RJ, Fachet M, Höffner K, Barton PI and Sundmacher K, Dynamic flux balance modeling to increase the production of high-value compounds in green microalgae. *Biotechnol Biofuels* **9**(1):165 (2016).
54. Wang L, Lai L, Ouyang Q and Tang C, Flux balance analysis of ammonia assimilation network in *E. coli* predicts preferred regulation point. *PLoS One* **6**(1):e16362 (2011).
55. Ng W, Effect of glucose and ammonium chloride supplementation and phosphate buffer on *Escherichia coli* DH5α growth in LB Lennox medium. *PeerJ* **9**:55–57 (2018).
56. Vlysidis A, Binns M, Webb C and Theodoropoulos C, Glycerol utilisation for the production of chemicals: Conversion to succinic acid, a combined experimental and computational study. *Biochem Eng J* **58–59**(1):1–11 (2011).
57. Wong KK and Kwan HS, Transcription of glpT of Escherichia coli K12 is regulated by anaerobiosis and fnr. *FEMS Microbiol Lett* **94**(1–2):15–18 (1992).
58. Lemieux MJ, Huang Y and Wang DN, Crystal structure and mechanism of GlpT, the glycerol-3-phosphate transporter from E. coli. *J Electron Microsc* **54**(Suppl. 1):43–46 (2005).
59. Bren A, Park JO, Towbin BD, Dekel E, Rabinowitz JD and Alon U, Glucose becomes one of the worst carbon sources for *E.coli* on poor nitrogen sources due to suboptimal levels of cAMP. *Sci Rep* **6**(1):24834 (2016).
60. Conrad TM, Frazier M, Joyce AR, Cho B-K, Knight EM, Lewis NE *et al*., RNA polymerase mutants found through adaptive evolution reprogram *Escherichia coli* for optimal growth in minimal media. *Proc Natl Acad Sci U S A* **107**(47):20500–20505 (2010).
61. Lyubetskaya AV, Rubanov LI and Gelfand MS, Use of the flux model of amino acid metabolism of *Escherichia coli*. *Biochemistry* **71**(11):1256–1260 (2006).

Albert Enrique Tafur Rangel

Albert Enrique Tafur Rangel is a microbiologist with a master's degree in computational biology and a PhD in engineering. His research focuses on the integration of multi-omics data with metabolic models and bioprocess for metabolic modification of biological systems, using bioinformatics and systems biology.



Luis H. Reyes

Luis H. Reyes is an associate professor at Universidad de Los Andes. His research interest consists in applying biological engineering to design and develop bioprocesses and bioproducts using several tools borrowed from the natural sciences such as molecular and synthetic biology, reverse engineering of microorganisms, adaptive laboratory evolution, and microbiology.

Jorge Mario Gómez Ramírez

Jorge Mario Gómez Ramírez is a professor at Universidad de Los Andes. His research aims to improve the efficiency of industrial operations, and to design chemical and food products with an optimal balance between utility, economy, and quality.

Andrés Fernando González Barrios

Andrés Fernando González Barrios is a professor at Universidad de Los Andes. His research is focused on systems biology and, specifically, product design and process improvement in industry using metabolism, protein, and protein engineering modeling strategies.